What can be done, however, is computing an approximation of the smallest enclosing sphere, starting with the center of mass of the cluster as initial guess. At the very high level: the approximation algorithm tries at each iteration to move the center towards the farthest point in the dataset; after each iteration, the maximum distance the point can move is shortened, limited by the span of the previous update, thus ensuring convergence.

We won't delve deeper into this algorithm in this context; the interested readers can read more about this method starting, for instance, with this article[20] by Fischer et al.

For now, we will move to another way we could improve the balancing of the tree, reducing the nodes overlap.

10.5.3 Improved Split Heuristic

We have seen, in section 10.3, that splitting and merging nodes, as well as "borrowing" points/children from siblings, can result in skewed clusters that requires bounding envelopes larger than necessary, and increases nodes overlap.

To counteract this effect, Kurniawati et al., in their work[21] on `Ss+-trees` introduce a new split heuristic that, instead of only partitioning points along the direction of maximum variance, tries to find the two groups such that each of them will collect the closest nearby points.

To achieve this result, a variant of the k-means clustering algorithm will be used, with two constraints: